A Note on Exact Closed-Form Solutions for Lamb's Problem with an Arbitrary Poisson's Ratio

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ABSTRACT

Lamb's Problem, a cornerstone in theoretical seismology, investigates the displacement of an elastic half-space due to surficial or buried point sources. Although the exact closed-form solution for surface displacements from surficial sources is well established, the solutions for buried sources in the 3D case have remained limited, either confined to integral forms or applicable only for certain Poisson's ratios. This study addresses this gap by generalizing the existing closed-form solutions to accommodate Poisson's ratio for buried sources. Our solutions are expressed in terms of elementary algebraic expressions, elliptic integrals, and Gauss hypergeometric functions. The derived solution is validated by numerical tests. With the newly derived solution, we also explore the properties of the leaking mode, which stems from the complex conjugate roots of the Rayleigh equation.

KEY POINTS

- The exact closed-form solution for Lamb's problem with an arbitrary Poisson's ratio is derived.
- The derived formulation is tested numerically and the corresponding code is available online.
- The properties of the leaking mode are explored with the newly derived formulation.

INTRODUCTION

Lamb's problem involves the displacement response of an elastic half-space resulting from a point source. Since the seminal work of Lamb (1904), one primary focus of theoretical seismology in the first half of the twentieth century was the pursuit of an exact analytical solution to Lamb's problem (Ben-Menahem, 1995). Nowadays, the study of Lamb's problem remains of theoretical significance (Feng and Zhang, 2018), providing a foundational solution for the boundary element method to study earthquake rupture dynamics (Richards, 1979; Zhang and Chen, 2006; Dineva *et al.*, 2019).

Following initial efforts (Lamb, 1904; Lapwood, 1949; Nakano, 1955), the Cagniard–de Hoop method, originally proposed by Cagniard (1939) and later modified by de Hoop (1960), has become the standard approach to tackle Lamb's problem. Comprehensive reviews on the history and available solutions of the Lamb's problem can be seen in Dineva *et al.* (2019) and Emami and Eskandari-Ghadi (2019a). In a recent study, Feng and Zhang (2020b) classified 3D Lamb's problems into three types according to the locations of the source and the receiver.

The Lamb's problem of type I deals with the case when both the source and the receiver are located at the surface. Pekeris (1955b) and Chao (1960) derived closed-form solutions for vertical and tangential point sources, respectively, focusing on a specific Poisson's ratio ($\nu = 0.25$). Mooney (1974) extended these results to include vertical loads for arbitrary Poisson's ratios. Finally, Richards (1979) and Kausel (2013) provided complete closed-form solutions for Lamb's problem of type I.

The Lamb's problem of type II and type III investigate scenarios where the source is buried, with the receiver located at the surface (type II) or beneath the surface (type III). Following early attempts to solve the Lamb's problem of types II and III (Lapwood, 1949; Pinney, 1954; Pekeris, 1955a), Johnson (1974) presented a comprehensive integral form solution using the Cagniard–de Hoop technique. Detailed derivations can be found in Watanabe (2015) and Emami and Eskandari-Ghadi (2019b). Building upon the solution of Johnson (1974), Feng and Zhang (2018, 2020b) successfully derived closed-form solutions for Lamb's problem of types II and III, respectively. Employing a similar procedure, Feng and Zhang (2020a) also obtained an exact closed-form solution

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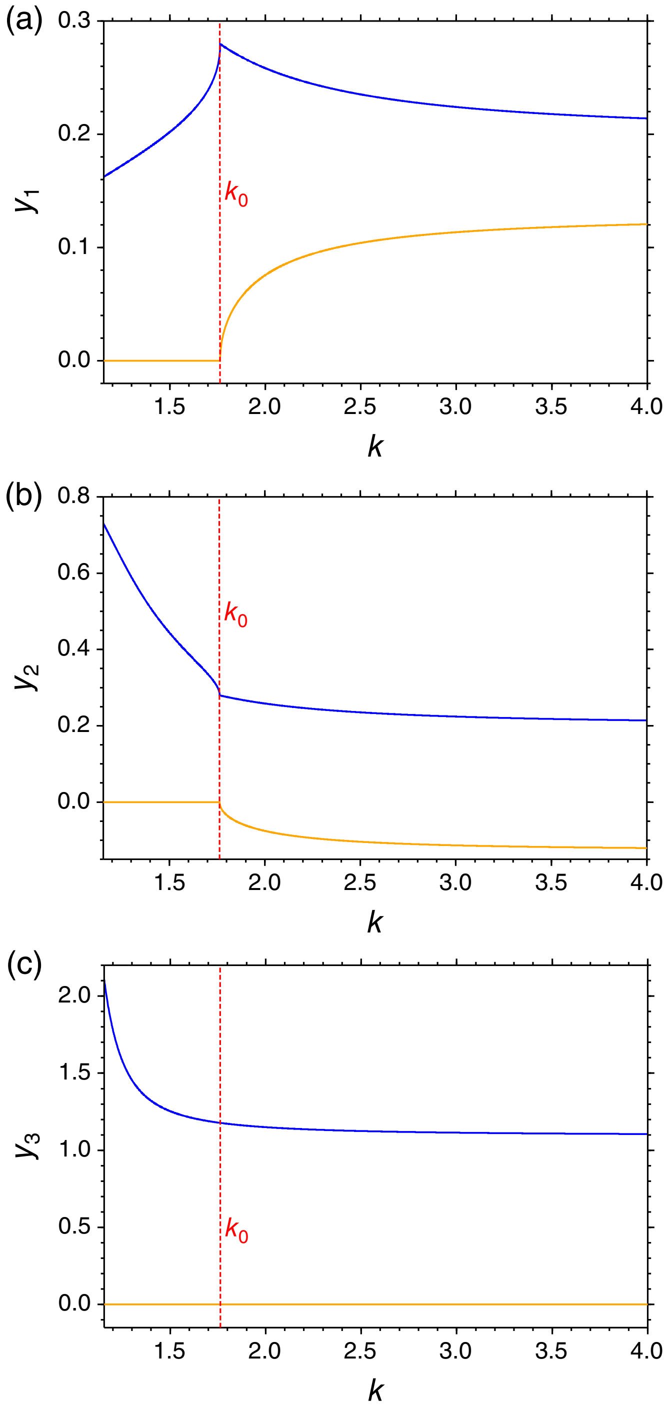

Figure 1. Three roots of the Rayleigh function (a) y_1, (b) y_2, and (c) y_3. Real and imaginary part of each root are in blue and yellow colors, respectively. y_3 is always real, corresponding to the Rayleigh wave. When $k > k_0$, y_1 and y_2 become complex, $\sim k_0 = 1.7636$. The color version of this figure is available only in the electronic edition.

for a moving source. The closed-form solution is exclusively composed of elementary algebraic functions and elliptic integrals, offering advantages over integral form solutions by enabling specific wave type analysis. However, the formulations provided by Feng and Zhang (2018, 2020a,b) are only valid for certain Poisson's ratios ($0 < \nu < 0.2631$). In a recent study, based upon the principle of equipartition of energy, Sánchez-Sesma *et al.* (2023) expressed the imaginary part of Green's tensor in an elastic half-space in frequency domain in terms of cross correlations within a synthetic diffuse field composed by incident homogeneous plane *P*, *SV*, and *SH* waves, with incident angles uniformly distributed in the associated half-sphere, including their reflected consequences, and azimuthally distributed plane Rayleigh waves.

When the Poisson's ratio (ν) exceeds 0.2631, Rayleigh's equation exhibits two complex conjugate roots corresponding to the leaking mode (Schröder and Scott, 2001). The leaking mode is associated with the *PL* wave in the time domain (Oliver and Major, 1960; Lodge *et al.*, 1999). Recent advancements in the extraction of the leaking mode in observational seismology (Li *et al.*, 2021, 2022) have led to increased interest in understanding its properties (Kennett, 2023). Chapman (1972) investigated and examined the significance of the leaking mode in Lamb's problem in the 2D case. The utilization of a closed-form solution for Lamb's problem enables a detailed exploration of the leaking mode's properties, especially in the 3D case.

In this study, by carefully examining the exact closed-form solution of Feng and Zhang (2018) for Lamb's problem of type II, we successfully extended their solution to be applicable for large Poisson's ratios ($\nu > 0.2631$). By integrating the results of Feng and Zhang (2018), we have made the complete closed-form solution for Lamb's problem of type II available. Using the newly derived solution, we conducted an analysis of the contribution of the leaking mode to the Green's function. In addition, employing a similar approach, the exact closed-form solution of the Lamb's problem of type III by Feng and Zhang (2020a) and for moving sources Feng and Zhang (2020b) can potentially be extended to arbitrary Poisson ratios.

THEORY

In this section, we begin by analyzing the roots of the Rayleigh equation across various Poisson's ratio, demonstrating that only specific terms in Feng and Zhang (2018) require modification for large Poisson's ratios. Subsequently, we expound upon the methods employed to adapt these particular terms, rendering them suitable for a wide range of Poisson's ratios. Our derivation process adheres to the conventions of Feng and Zhang (2018).

Roots of the Rayleigh equation

The Rayleigh's function is as follows (Feng and Zhang, 2018, their equation 8):

$$R(y) = -16(1 - k^{-2})y^3 + 8(3 - 2k^{-2})y^2 - 8y + 1, \quad (1)$$

in which k is the V_P/V_S ratio, which is related to Poisson's ratio (ν) as $\nu = \frac{1}{2}\frac{k^2-2}{k^2-1}$. Because ν is in the range $(-1, \frac{1}{2})$, k is in the range $(\frac{2}{\sqrt{3}}, \infty)$ correspondingly.

The Rayleigh equation, denoted by $R(y) = 0$, yields three roots. The three roots can be graphically represented as a function a function of k in Figure 1. It is shown y_3 is always positive, corresponding to the Rayleigh wave. Particularly, we note it is always true that $y_3 > 1$. The other two roots y_1 and y_2 are real when $k < k_0$, and are complex when $k > k_0$. k_0 is the real solution of the equation $11k_0^6 - 62k_0^4 + 107k_0^2 - 64 = 0$, which can be solved accurately using the root formula for the cubic equation. Approximately, $k_0 = 1.7636$, corresponding to the Poisson's ratio $\nu_0 = 0.2631$.

Feng and Zhang (2018) discussed the case $k < k_0$ when the roots of the Rayleigh function are all real (Fig. 1). When $k > k_0$, two of the roots become complex, and that is where the complexity is introduced to the exact closed-form solution for large Poisson's ratios.

Exact closed-form solution for Lamb's problem

The full Green's function can be decomposed as follows (Feng and Zhang, 2018, their equation 1):

$$\mathbf{G} = \mathbf{G}^P + \mathbf{G}^S + \mathbf{G}^{S-P}. \tag{2}$$

We first consider $\mathbf{G}^P$, and then deal with other terms.

By variable substitution and expansion of a rational function, it has been shown that $\mathbf{G}^P$ can be expressed as follows (Feng and Zhang, 2018, their equation 28):

$$\begin{aligned} G_{ij}^{\mathrm{P}} = \frac{1}{\pi^2 \mu r}\frac{\partial}{\partial t}\{ & H(t-t_{\mathrm{P}})\Big[\boxed{u_{ij,1}^{\mathrm{P}}U_1^{\mathrm{P}}(a_1^{\mathrm{P}}) + u_{ij,1}^{\mathrm{P}}U_1^{\mathrm{P}}(-a_1^{\mathrm{P}}) + u_{ij,1}^{\mathrm{P}}U_1^{\mathrm{P}}(a_2^{\mathrm{P}}) + u_{ij,1}^{\mathrm{P}}U_1^{\mathrm{P}}(-a_2^{\mathrm{P}})} \\ & + u_{ij,5}^{\mathrm{P}}U_2^{\mathrm{P}}(a_3^{\mathrm{P}}) + u_{ij,6}^{\mathrm{P}}U_3^{\mathrm{P}}(a_3^{\mathrm{P}}) + u_{ij,7}^{\mathrm{P}}U_4^{\mathrm{P}} + u_{ij,8}^{\mathrm{P}}U_5^{\mathrm{P}} + u_{ij,9}^{\mathrm{P}}U_6^{\mathrm{P}}\Big], \\ & H(t-t_{\mathrm{P}})\Big[\boxed{v_{ij,1}^{\mathrm{P}}V_1^{\mathrm{P}}(a_1^{\mathrm{P}}) + v_{ij,1}^{\mathrm{P}}V_1^{\mathrm{P}}(-a_1^{\mathrm{P}}) + v_{ij,1}^{\mathrm{P}}V_1^{\mathrm{P}}(a_2^{\mathrm{P}}) + v_{ij,1}^{\mathrm{P}}V_1^{\mathrm{P}}(-a_2^{\mathrm{P}})} \\ & + v_{ij,5}^{\mathrm{P}}V_2^{\mathrm{P}}(a_3^{\mathrm{P}}) + v_{ij,6}^{\mathrm{P}}V_3^{\mathrm{P}}(a_3^{\mathrm{P}}) + v_{ij,7}^{\mathrm{P}}V_4^{\mathrm{P}} + v_{ij,8}^{\mathrm{P}}V_5^{\mathrm{P}} + v_{ij,9}^{\mathrm{P}}V_6^{\mathrm{P}} + v_{ij,10}^{\mathrm{P}}V_7^{\mathrm{P}}\Big] \end{aligned} \tag{3}$$

in which meanings of the variables can be seen in Feng and Zhang (2018). Because $(a_1^P)^2 = 1 - k^2 y_1$, $(a_2^P)^2 = 1 - k^2 y_2$, and $(a_3^P)^2 = k^2 y_3 - 1$, a_1^P and a_2^P will be complex for $k > k_0$, and a_3^P will always be real because $k_3 > 1$. Therefore, for a large V_P/V_S ratio, the terms enclosed in the box in equation (3) are the only ones affected. All other terms will be consistent with those derived in Feng and Zhang (2018).

In the following, we show how to compute $U_1^P(a)$ and $V_1^P(a)$ for a complex a.

Calculation of $U_1^P(a)$ for a complex a

$U_1^P(a)$ is expressed as follows:

$$U_1^P(a) = \mathrm{Re}\int_0^{\frac{\pi}{2}} \frac{\mathrm{d}x}{(T_{\mathrm{P}}\cos\theta - a) + \left(\mathrm{i}\sqrt{T_{\mathrm{P}}^2 - 1}\sin\theta\right)\cos x}. \tag{4}$$

The integral can be calculated using the Mooney's integral in the complex case, as proposed by Mooney (1974) and Kausel (2013).

The Mooney's integral in the complex case is expressed as follows (Mooney, 1974; Kausel, 2013):

$$\frac{2}{\pi}\int_0^{\frac{\pi}{2}} \frac{C\mathrm{d}\theta}{C + D\sin^2\theta} = \frac{\sqrt{Z}}{\sqrt{Z+1}}, \tag{5}$$

in which Z is complex, $Z = C/D$, and $Z(Z+1) \neq 0$.

Therefore, using the relation $\cos x = 1 - 2\sin^2\frac{x}{2}$ and assuming $y = \frac{x}{2}$,

$$U_1^P(a) = \mathrm{Re}\int_0^{\frac{\pi}{2}} \frac{\mathrm{d}y}{(T_P\cos\theta - a + \mathrm{i}\sqrt{T_P^2 - 1}\sin\theta) + (-2\mathrm{i}\sqrt{T_P^2 - 1}\sin\theta)\sin^2 y}. \tag{6}$$

$U_1^P(a)$ can then be computed as follows:

$$U_1^P(a) = \frac{\pi}{2}\mathrm{Re}\left\{\frac{1}{C}\int_0^{\frac{\pi}{2}} \frac{C\mathrm{d}y}{C + D\sin^2 y}\right\} = \frac{\pi}{2}\mathrm{Re}\left\{\frac{1}{C}\sqrt{\frac{C}{C+D}}\right\}, \tag{7}$$

in which

$$C = T_P\cos\theta - a + \mathrm{i}\sqrt{T_P^2 - 1}\sin\theta, \tag{8}$$

and

$$D = -2i\sqrt{T_P^2 - 1}\sin\theta. \tag{9}$$

It can be shown that equation (7) reduces to the form of equation (29) in Feng and Zhang (2018) when a is real.

Calculation of $V_1^P(a)$ for a complex a

$V_1^P(a)$ is expressed as follows:

$$V_1^P(a) = \mathrm{Im}\left\{\frac{1}{\sqrt{\xi_2(\xi_2 - T_P\cos\theta)}}\int_{\Gamma_1} \frac{1}{\frac{\xi_2 C - \xi_1}{C-1} - a}\frac{\mathrm{d}C}{\sqrt{[C^2 + (C_1^P)^2][C^2 + (C_2^P)^2]}}\right\}. \tag{10}$$

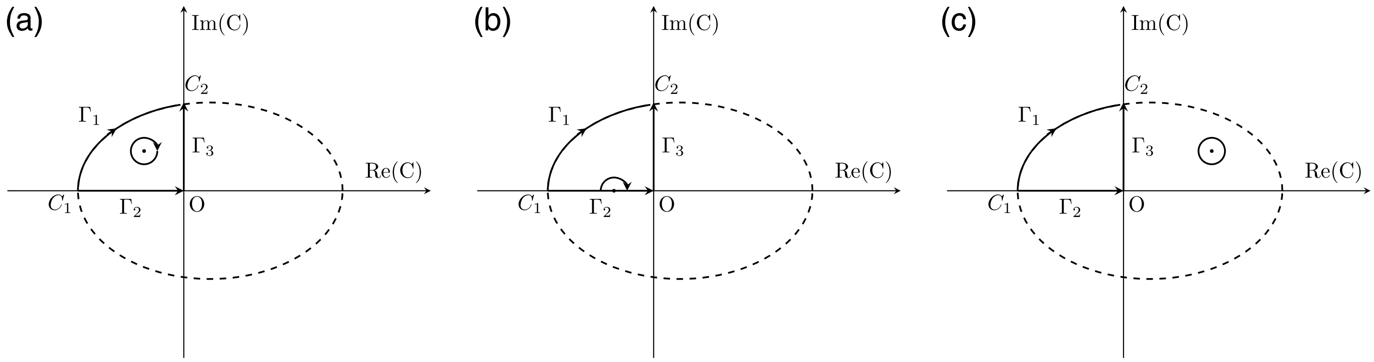

Figure 2. Integration path Γ_1 in equation (10), integration paths Γ_2 and Γ_3 in equation (12), and the location of pole that may contribute to the integration in equation (10). The pole may contribute to the integration differently according to its location. (a) The pole is located within the area defined by Γ_1, Γ_2, and Γ_3, the pole contribution is described in equation (14). (b) The pole is located along the x-axis, the pole contribution is discussed in equation (A12), and has been included in the solution of Feng and Zhang (2018). (c) The pole is located somewhere else, the pole does not contribute the integration.

The integral path Γ_1 is in the second quadrant along an ellipse in the complex domain as shown in Figure 2:

$$\left(\frac{x-1}{b}+1\right)^2+\frac{y^2}{c^2}=1. \tag{11}$$

Details of the derivation and the explicit expression of b and c are provided in Appendix A.

Utilizing the residue theorem of complex integration, equation (10) can be decomposed as follows:

$$\int_{\Gamma_1}=\int_{\Gamma_2+\Gamma_3}+V_1^{\mathrm{PI}}(a), \tag{12}$$

in which $V_1^{\mathrm{PI}}(a)$ denotes the pole contribution. The value of $V_1^{\mathrm{PI}}(a)$ depends on the location of the pole (Fig. 2). The scenario depicted in Figure 2b, in which the pole is located at the x-axis, has been previously discussed in Feng and Zhang (2018). In this study, our primary emphasis is directed toward the case in which the pole is positioned within the enclosed region delineated by Γ_1, Γ_2, and Γ_3 (Fig. 2a). Conversely, in the case presented in Figure 2c, $V_1^{\mathrm{PI}}=0$.

After some algebra manipulation, the integral in the case of Figure 2a can be further decomposed as follows:

$$V_1^P(a)=V_1^{\mathrm{PI}}(a)+\mathrm{Im}\{M_2^P V_1^{\mathrm{PII}}(a)+M_3^P V_1^{\mathrm{PIII}}(a)+M_4^P V_1^{\mathrm{PIV}}(a)\}, \tag{13}$$

in which

$$V_1^{\mathrm{PI}}(a)=-\mathrm{Im}\left\{\frac{2\pi i}{\sqrt{(a^2+T_P^2-2aT_P\cos\theta-\sin^2\theta)(a^2+k^2-1)}}\right\}, \tag{14}$$

$$V_1^{\mathrm{PII}}(a)=-\frac{i}{C_2^P}K\left(\frac{C_1^P}{C_2^P},\frac{iC_1}{C_1^P}\right)+\frac{i}{C_1^P}K\left(\frac{C_2^P}{C_1^P},1\right), \tag{15}$$

$$V_1^{\mathrm{PIII}}(a)=\frac{1}{2}F^P(x)\Big|_{(C_1^P)^2}^{(C_2^P)^2}, \tag{16}$$

$$V_1^{\mathrm{PIV}}(a)=-\frac{i}{C_3C_2^P}\Pi\left(-\left(\frac{C_1^P}{C_3}\right)^2,\frac{C_1^P}{C_2^P},\frac{iC_1}{C_1^P}\right) -\frac{i}{C_3C_1^P}\Pi\left(-\left(\frac{C_2^P}{C_3}\right)^2,\frac{C_2^P}{C_1^P},1\right), \tag{17}$$

$M_2^P=\frac{1}{\xi_2(\xi_2-T_P\cos\theta)}\frac{1}{\xi_2-a}$, $M_3^P=M_2^P\frac{\xi_1-\xi_2}{\xi_2-a}$, $M_4^P=-M_3^P$, $C_3=-\frac{\xi_1-a}{\xi_2-a}$, and the explicit expression of F^P is discussed in Appendix B and Appendix C. The incomplete elliptic integrals $K(\tau,y)$ and $\Pi(c,\tau,y)$ are used in the above expression, which are defined as follows:

$$K(\tau,y)=\int_0^y\frac{\mathrm{d}x}{\sqrt{1-x^2}\sqrt{1-\tau^2x^2}}, \tag{18}$$

and

$$\Pi(c,\tau,y)=\int_0^y\frac{1}{1-cx^2}\frac{\mathrm{d}x}{\sqrt{1-x^2}\sqrt{1-\tau^2x^2}}. \tag{19}$$

The explicit expression of F^P is discussed in Appendix C. Detailed derivation of $V_1^{\mathrm{PI}}(a)$ can be found in Appendix A, and detailed derivation of other terms is in Appendix B.

Calculation of $U_1^S(a)$, $V_1^{S-P}(a)$, $V_1^S(a)$ for a complex a

$U_1^S(a)$ has the same closed-form as in equation (7), except that T_P should be replaced by T_S.

The computations of $V_1^{S-P}(a)$ and $V_1^S(a)$ are relatively complex. Similar as $V_1^P(a)$, $V_1^{S-P}(a)$ can be decomposed as follows:

$$V_1^{S-P}(a)=-\mathrm{Im}\{M_2^{S-P}V_1^{S-P,\mathrm{II}}(a)+M_3^{S-P}V_1^{S-P,\mathrm{III}}(a) +M_4^{S-P}V_1^{S-P,\mathrm{IV}}(a)\}, \tag{20}$$

in which

$$V_1^{S-P,\mathrm{II}}(a) = \frac{i}{C_1^S} K\left(\frac{C_2^S}{C_1^S}, \frac{iC_2}{C_2^S}\right) - \frac{i}{C_1^S} K\left(\frac{C_2^S}{C_1^S}, \frac{iC_1}{C_2^S}\right), \tag{21}$$

$$V_1^{S-P,\mathrm{III}}(a) = \frac{1}{2} F^S(x)\Big|_{(C_1^S)^2}^{(C_2^S)^2}, \tag{22}$$

$$\begin{aligned} V_1^{S-P,\mathrm{IV}}(a) = &-\frac{i}{C_3 C_1^S}\Pi\left(-\left(\frac{C_2^S}{C_3}\right)^2, \frac{C_2^S}{C_1^S}, \frac{iC_2}{C_2^S}\right) \\ &+ \frac{i}{C_3 C_1^S}\Pi\left(-\left(\frac{C_2^S}{C_3}\right)^2, \frac{C_2^S}{C_1^S}, \frac{iC_1}{C_2^S}\right), \end{aligned} \tag{23}$$

$$\begin{aligned} &M_2^{S-P} = \frac{1}{\xi_2(\xi_2 - T_S\cos\theta)} \frac{1}{\xi_2 - a}, \\ &M_3^{S-P} = M_2^{S-P}\frac{\xi_1 - \xi_2}{\xi_2 - a}, \text{and } M_4^{S-P} = -M_3^{S-P}. \end{aligned} \tag{24}$$

To compute $V_1^S(a)$, we use $\xi_2 > \xi_1$ to ensure all the coefficients are real. Therefore,

$$\begin{aligned} V_1^S(a) == &V_1^{S,1\mathrm{I}}(a) + \mathrm{Im}\{M_2^S V_1^{S,1\mathrm{II}}(a) + M_3^S V_1^{S,1\mathrm{III}}(a) \\ &+ M_4^S V_1^{S,1\mathrm{IV}}(a)\} - H\left(T_S\cos\theta - \sqrt{1-\frac{1}{k^2}}\right)\mathrm{Im}\{M_2^S V_1^{S,2\mathrm{II}}(a) \\ &+ M_3^S V_1^{S,2\mathrm{III}}(a) + M_4^S V_1^{S,2\mathrm{IV}}(a)\}, \end{aligned} \tag{25}$$

in which

$$V_1^{S,1\mathrm{I}}(a) = \mathrm{Im}\left\{\frac{2\pi i}{\sqrt{(a^2 + T_S^2 - 2aT_S\cos\theta - \sin^2\theta)(a^2 + 1/k^2 - 1)}}\right\}, \tag{26}$$

$$V_1^{S,1\mathrm{II}}(a) = \frac{i}{C_2^S} K\left(\frac{C_1^S}{C_2^S}, \frac{iC_1^{(1)}}{C_1^S}\right) + \frac{i}{C_1^S} K\left(\frac{C_2^S}{C_1^S}, 1\right), \tag{27}$$

$$V_1^{S,1\mathrm{III}}(a) = \frac{1}{2} F^S(x)\Big|_{(C_1^{(1)})^2}^{(C_2^{(1)})^2}, \tag{28}$$

$$\begin{aligned} V_1^{S,1\mathrm{IV}}(a) = &-\frac{i}{C_3^2 C_2^S}\Pi\left(-\left(\frac{C_1^S}{C_3}\right)^2, \frac{C_1^S}{C_2^S}, \frac{iC_1^{(1)}}{C_1^S}\right) \\ &- \frac{i}{C_3^2 C_1^S}\Pi\left(-\left(\frac{C_2^S}{C_3}\right)^2, \frac{C_2^S}{C_1^S}, 1\right), \end{aligned} \tag{29}$$

$$V_1^{S,2\mathrm{II}}(a) = \frac{i}{C_1^S} K\left(\frac{C_2^S}{C_1^S}, -\frac{iC_2^{(2)}}{C_2^S}\right) - \frac{i}{C_1^S} K\left(\frac{C_2^S}{C_1^S}, -\frac{iC_1^{(2)}}{C_2^S}\right), \tag{30}$$

$$V_1^{S,2\mathrm{III}}(a) = \frac{1}{2} F^S(x)\Big|_{\left(C_1^{(2)}\right)^2}^{\left(C_2^{(2)}\right)^2}, \tag{31}$$

$$\begin{aligned} V_1^{S,2\mathrm{IV}}(a) = &-\frac{i}{C_3^2 C_1^S}\Pi\left(-\left(\frac{C_2^S}{C_3}\right)^2, \frac{C_2^S}{C_1^S}, -\frac{iC_2^{(2)}}{C_2^S}\right) \\ &+ \frac{i}{C_3^2 C_1^S}\Pi\left(-\left(\frac{C_2^S}{C_3}\right)^2, \frac{C_2^S}{C_1^S}, -\frac{iC_1^{(2)}}{C_2^S}\right), \end{aligned} \tag{32}$$

in which $M_2^S = -\frac{1}{\xi_2(\xi_2 - T_S\cos\theta)}\frac{1}{\xi_2 - a}$, $M_3^S = M_2^S\frac{\xi_1 - \xi_2}{\xi_2 - a}$, $M_4^S = -M_3^S C_3$, $C_1^{(1)} = -\frac{\xi_1 - T_S\cos\theta}{T_S\cos\theta - \xi_2}$, $C_2^{(1)} = i\sqrt{C_1^{(1)}}$, $C_1^{(2)} = \frac{T_S\cos\theta - \xi_1}{T_S\cos\theta - \xi_2}$, and $C_2^{(2)} = \frac{\sqrt{1-1/k^2} - \xi_1}{\sqrt{1-1/k^2} - \xi_2}$. The explicit expression of F^S is discussed in Appendix C.

NUMERICAL RESULTS

The code from Feng and Zhang (2018) can be extended to accommodate arbitrary Poisson's ratios by adjusting specific functions associated with $U_1^P(a)$, $V_1^P(a)$, $U_1^S(a)$, $V_1^S(a)$, and $V_1^{S-P}(a)$ within their package. It is noteworthy that the extended version of the exact closed-form solution for large Poisson's ratios includes both complete and incomplete elliptical integrals as well as the Gaussian hypergeometric function.

In this section, we validate the modified code using a half-space model with the parameters P-wave velocity $V_P = 8.00$ km/s, V_P/V_S ratio $k = 3$, and a density $\rho = 3.30$ g/cm^3. The epicenter distance is chosen to be 10 km. Following Johnson (1974) and Feng and Zhang (2018), the symbol $G_{ij}(x_1, x_2, x_3, t; x_1', x_2', x_3', \tau)$ is used to denote the Green's function, in which $x(x_1, x_2, x_3)$ is the receiver location, $x'(x_1', x_2', x_3')$ is the source location. In this study, our results are compared with the numerical integration result of Johnson (1974) (Fig. 3). Specifically, we compare the outcome of the closed-form solution and numerical integral method for $G_{ij}(10,0,0,t;0,0,0.2,0)$ (Fig. 3a) and $G_{ij}(10,0,0,t;0,0,2,0)$ (Fig. 3b), respectively. The agreements between these results validate the accuracy and reliability of our code.

Next, we separate G_{ij} into two parts:

$$G_{ij} = L_{ij} + \bar{O}_{ij}, \tag{33}$$

in which

$$\begin{aligned} L_{ij} = &H(t - t_P)[u_{ij,1}^P U_1^P(a_1^P) + u_{ij,2}^P U_1^P(-a_1^P) + u_{ij,3}^P U_1^P(a_2^P) \\ &+ u_{ij,4}^P U_1^P(-a_2^P) + v_{ij,1}^P V_1^P(a_1^P) + v_{ij,2}^P V_1^P(-a_1^P) + v_{ij,3}^P V_1^P(a_2^P) \\ &+ v_{ij,4}^P V_1^P(-a_2^P)] + H(t - t_S)[u_{ij,1}^S U_1^S(a_1^S) + u_{ij,2}^S U_1^S(-a_1^S) \\ &+ u_{ij,3}^S U_1^S(a_2^S) + u_{ij,4}^S U_1^S(-a_2^S) + v_{ij,1}^S V_1^S(a_1^S) + v_{ij,2}^S V_1^S(-a_1^S) \\ &+ v_{ij,3}^S V_1^S(a_2^S) + v_{ij,4}^S V_1^S(-a_2^S)] + H\left(\sin\theta - \frac{1}{k}\right)\left[v_{ij,1}^S V_1^{S-P}(a_1^S)\right. \\ &\left. + v_{ij,2}^S V_1^{S-P}(-a_1^S) + v_{ij,3}^S V_1^{S-P}(a_2^S) + v_{ij,4}^S V_1^{S-P}(-a_2^S)\right], \end{aligned} \tag{34}$$

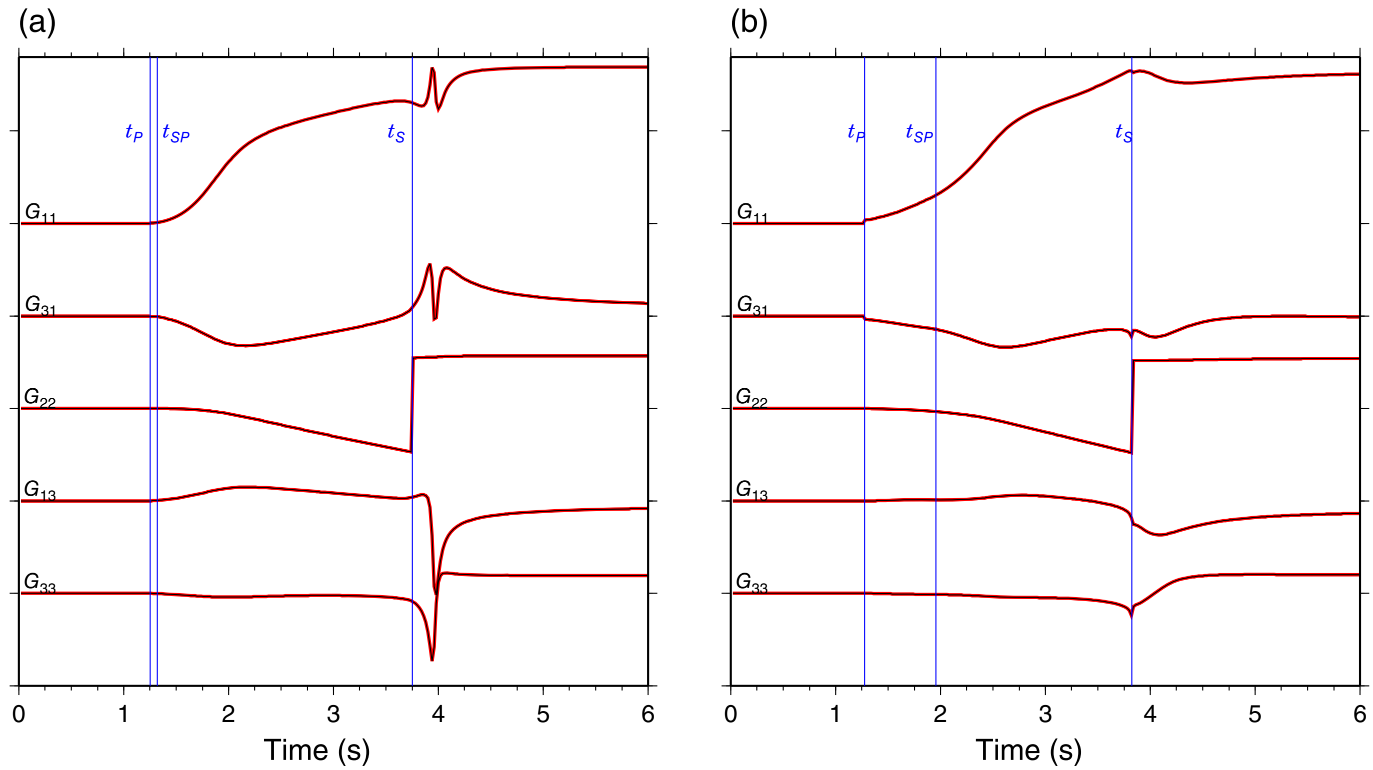

Figure 3. Comparison between Johnson (1974) solution and our solution for all nonzero components. (a) Comparison of $G_{ij}(10,0,-0,t;0,0,0.2,0)$ between the two formulation. (b) Comparison of $G_{ij}(10,0,-0,t;0,0,2,0)$ between the two formulation. Johnson (1974) results are in red, and our results are in black. The arrival times of t_P, t_{SP}, and t_S are plotted and labeled. The color version of this figure is available only in the electronic edition.

and

$$\bar{O}_{ij} = G_{ij} - L_{ij}. \tag{35}$$

L_{ij} is identified as the term that encompasses the contribution from the complex conjugate roots of the Rayleigh function, and, thus, it can be referred to as the leaking mode term. We use $\bar{O}_{ij}$ to represent the remaining portion of the full Green's function, exclusive of the leaking mode.

We have plotted the leaking mode term L_{ij} alongside the whole solution G_{ij} for various source depths ranging from 1 to 10 km (Fig. 4). Chapman (1972) concluded that, in a 2D scenario, the leaking mode may either result in a distinct arrival or influence the shape of other arrivals. Our numerical tests in a 3D context further reveal that the leaking mode term significantly contributes to the phase near t_{S-P}, and also affects the P arrival at t_P and S arrival at t_S (Fig. 4).

In the review process of this article, we realized that a monograph with a comprehensive investigation of the exact closed-form solution of the Lamb's problem (Zhang and Feng, 2024) has recently been published. We would like to point out that the formulas presented by Zhang and Feng (2024) are in different form from ours. In principle, the formulas of Zhang and Feng (2024) would be equivalent to ours in the mathematical point of view. Although detailed comparison between both sets of the formulations would be interesting, it is beyond the scope of this study.

CONCLUSION

In this study, we re-examine the precise closed-form solution for Lamb's problem, with a specific focus on scenarios in which the source is buried and the receiver is located at the surface. We extend the solution provided by Feng and Zhang (2018) to accommodate an arbitrary Poisson's ratio. The extended solution comprises algebraic expressions, complete and incomplete elliptic functions, and the Gaussian hypergeometric function. Through numerical comparisons with the solution of Johnson (1974) in integral form, we verify the accuracy of our formulation for large Poisson's ratios. Utilizing this formulation, we explore the contribution of the leaking mode to the full Green's function.

DATA AND RESOURCES

There are no new data associated with this article. The Matlab scripts (www.mathworks.com/products/matlab, last accessed January 2024) used in this study, modified from Feng and Zhang (2018), are available online (https://github.com/ShaoqianHu/lamb_large_poisson_ratio, last accessed April 2024).

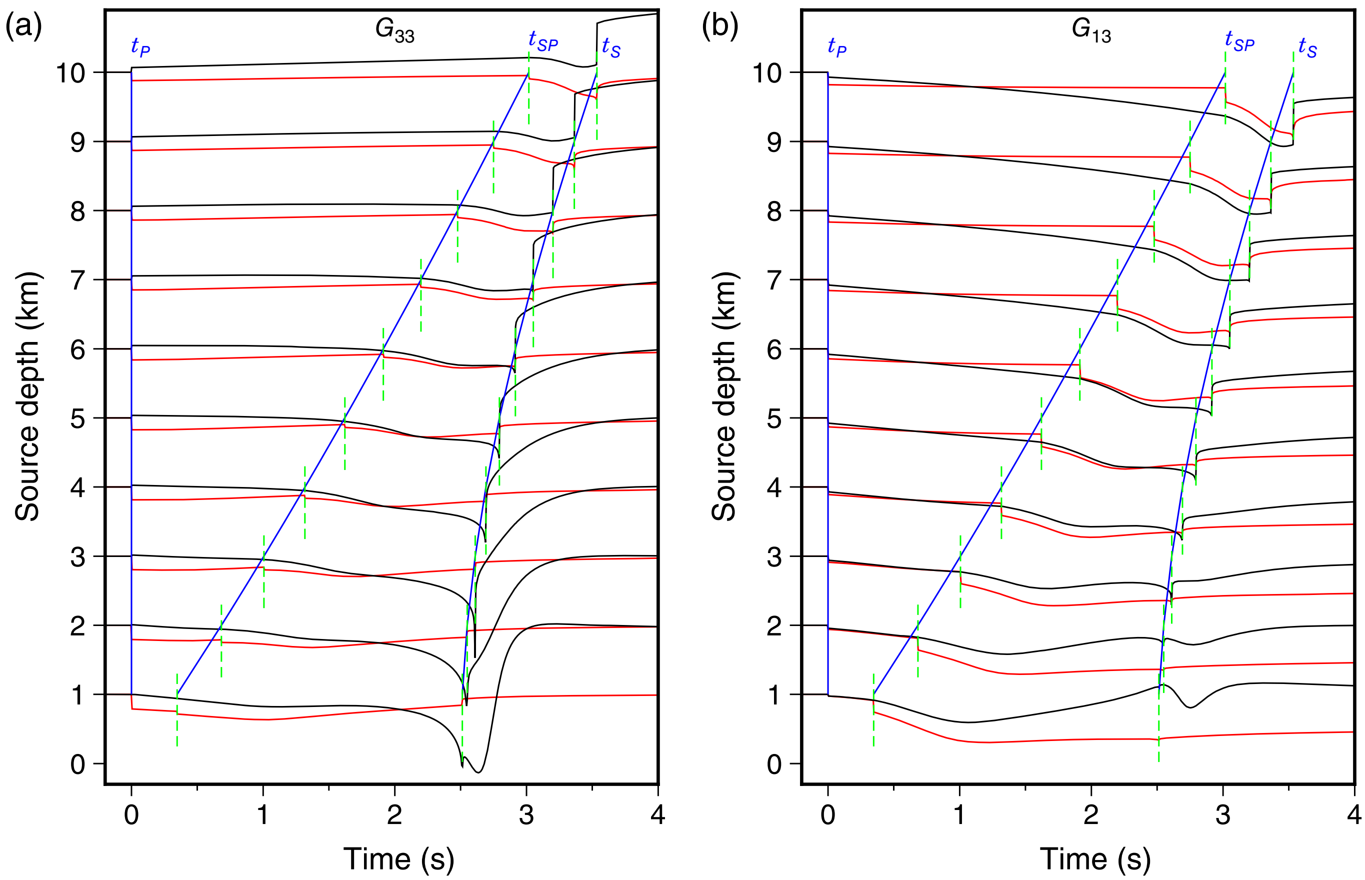

Figure 4. The effects of leaking mode term L_{ij} on the full Green's function G_{ij} for different source depths (1–10 m). L_{ij} is in red, and G_{ij} is in black. All traces are aligned with the P-wave arrival t_P. The relative arrival times of t_{S-P} and t_S are plotted and labeled. (a) G_{33} component. (b) G_{13} component. The color version of this figure is available only in the electronic edition.

DECLARATION OF COMPETING INTERESTS

The authors acknowledge that there are no conflicts of interest recorded.

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APPENDIX A

Derivation of $V_1^{PI}(A)$

We first determine the explicit equation of the path Γ_1.

Because $B = \frac{\xi_2 C - \xi_1}{C-1}$, inversely, there is a relation,

$$C = \frac{B - \xi_1}{B - \xi_2}. \tag{A1}$$

Because B is in the range $[T_P \cos\theta, T_P \cos\theta + i\sqrt{T_P^2 - 1}\sin\theta]$, B can be parameterized as follows:

$$B = T_P \cos\theta + \left(i\sqrt{T_P^2 - 1}\sin\theta\right)t, \tag{A2}$$

in which t is an auxiliary parameter, ranging from 0 to 1.

Therefore, substituting equation (A2) into equation (A1), we have the following equation:

$$\mathrm{Re}\{C\} = \frac{(t^2 - 1)(T_P^2 - 1)\sin^2\theta}{(T_P\cos\theta - \xi_2)^2 + t^2(T_P^2 - 1)\sin^2\theta}, \tag{A3}$$

and

$$\mathrm{Im}\{C\} = \frac{t(\xi_1 - \xi_2)\sqrt{T_P^2 - 1}\sin\theta}{(T_P\cos\theta - \xi_2)^2 + t^2(T_P^2 - 1)\sin^2\theta}. \tag{A4}$$

After performing algebraic manipulations, the real ($\mathrm{Re}\{C\}$) and imaginary ($\mathrm{Im}\{C\}$) parts of C, denoted as x and y, respectively, for brevity, are found to lie on an ellipse in the complex coordinate system (Fig. 2),

$$\left(\frac{x-1}{b} + 1\right)^2 + \frac{y^2}{c^2} = 1, \tag{A5}$$

in which

$$b = \frac{(T_P^2 - 1)\sin^2\theta}{2(T_P\cos\theta - \xi_2)^2} + \frac{1}{2}, \tag{A6}$$

and

$$c = \frac{\xi_1 - \xi_2}{2(T_P\cos\theta - \xi_2)}. \tag{A7}$$

Next, we define the following equation:

$$f(C) = \frac{1}{\xi_2 - a}\frac{C-1}{C - \frac{\xi_1 - a}{\xi_2 - a}}\frac{1}{\sqrt{[C^2 + (C_1^P)^2][C^2 + (C_2^P)^2]}}, \tag{A8}$$

in the following analysis.

For the case as shown in Figure 2a,

$$V_1^{\mathrm{PI}}(a) = -\mathrm{Im}\left\{i\pi\mathrm{Res}\left(f, \frac{\xi_1 - a}{\xi_2 - a}\right)\right\}. \tag{A9}$$

For the case in Figure 2b,

$$V_1^{\mathrm{PI}}(a) = -\mathrm{Im}\left\{2i\pi\mathrm{Res}\left(f, \frac{\xi_1 - a}{\xi_2 - a}\right)\right\}. \tag{A10}$$

In other cases, $V_1^{PI}(a) = 0$.

Because

$$\mathrm{Res}\left(f, \frac{\xi_1 - a}{\xi_2 - a}\right) = \frac{1}{\sqrt{(a^2 + T_P^2 - 2aT_P\cos\theta - \sin^2\theta)(a^2 + k^2 - 1)}}. \tag{A11}$$

In the case shown in Figure 2b,

$$V_1^{\mathrm{PI}}(a) = -\frac{\pi}{\sqrt{(a^2 + T_P^2 - 2aT_P\cos\theta - \sin^2\theta)(a^2 + k^2 - 1)}}. \tag{A12}$$

That is what Feng and Zhang (2018) obtained in their equation (33).

In the case shown in Figure 2a,

$$V_1^{PI}(a) = -\mathrm{Im}\left\{\frac{2\pi i}{\sqrt{(a^2 + T_P^2 - 2aT_P\cos\theta - \sin^2\theta)(a^2 + k^2 - 1)}}\right\}. \tag{A13}$$

APPENDIX B

Derivation of $V_1^{\mathrm{PII}}(a)$, $V_1^{\mathrm{PII}}(a)$, and $V_1^{\mathrm{PII}}(a)$

$V_1^P(a)$ in equation (10) can be decomposed as follows:

$$\begin{aligned} V_1^P(a) = \mathrm{Im}\Bigg\{ & M_2^P \int_{\Gamma_1} \frac{\mathrm{d}C}{\sqrt{[C^2 + (C_1^P)^2][C^2 + (C_2^P)^2]}} \\ & + M_3^P \int_{\Gamma_1} \frac{C\mathrm{d}C}{(C^2 - C_3^2)\sqrt{[C^2 + (C_1^P)^2][C^2 + (C_2^P)^2]}} \\ & + M_4^P \int_{\Gamma_1} \frac{\mathrm{d}C}{(C^2 - C_3^2)\sqrt{[C^2 + (C_1^P)^2][C^2 + (C_2^P)^2]}}\Bigg\}, \end{aligned} \tag{B1}$$

in which

$$M_2^P = \frac{1}{\xi_2(\xi_2 - T_P\cos\theta)}\frac{1}{\xi_2 - a}, \tag{B2}$$

$$M_3^P = M_2^P\frac{\xi_1 - \xi_2}{\xi_2 - a}, \tag{B3}$$

$$M_4^P = -M_2^P C_3 \frac{\xi_1 - \xi_2}{\xi_2 - a}, \tag{B4}$$

and

$$C_3 = -\frac{\xi_1 - a}{\xi_2 - a}. \tag{B5}$$

After changing the integration path as shown in equation (14), we have equation (15), in which V_1^{PI} is derived in Appendix A, and

$$V_1^{\mathrm{PII}}(a) = \int_{\Gamma_2+\Gamma_3} \frac{\mathrm{d}C}{\sqrt{[C^2 + (C_1^P)^2][C^2 + (C_2^P)^2]}}, \tag{B6}$$

$$V_1^{\mathrm{PIII}}(a) = \int_{\Gamma_2+\Gamma_3} \frac{C}{C^2 - C_3^2}\frac{\mathrm{d}C}{\sqrt{[C^2 + (C_1^P)^2][C^2 + (C_2^P)^2]}}, \tag{B7}$$

$$V_1^{\mathrm{PIV}}(a) = \int_{\Gamma_2+\Gamma_3} \frac{1}{C^2 - C_3^2}\frac{\mathrm{d}C}{\sqrt{[C^2 + (C_1^P)^2][C^2 + (C_2^P)^2]}}. \tag{B8}$$

1. V_1^{PII} Along the path Γ_2, $C_1 = \frac{\xi_1 - T_P\cos\theta}{T_P\cos\theta - \xi_2} \neq C_1^P$. Use the substitution $C = iC_1^P y$, we have the following equation:

$$\begin{aligned} V_1^{\mathrm{PII}}(a;\Gamma_2) &= \frac{i}{C_2^P}\int_0^{i\frac{C_1}{C_1^P}} \frac{\mathrm{d}y}{\sqrt{\left(1 - y^2\right)\left[1 - \left(\frac{C_1^P}{C_2^P}\right)^2 y^2\right]}} \\ &= \frac{i}{C_2^P}K\left(\frac{C_1^P}{C_2^P}, i\frac{C_1}{C_1^P}\right). \end{aligned} \tag{B9}$$

Along the path Γ_3, $C_2 = \sqrt{\frac{\xi_1 - T_P cos\theta}{T_P cos\theta - \xi_2}} = iC_2^P$. Use the substitution $C = iC_2^P y$, we have the following equation:

$$V_1^{\mathrm{PII}}(a;\Gamma_3) = \frac{i}{C_1^P}\int_0^1 \frac{\mathrm{d}y}{\sqrt{1 - y^2}\sqrt{1 - \left(\frac{C_2^P}{C_1^P}\right)^2 y^2}} = \frac{i}{C_2^P}K\left(\frac{C_2^P}{C_1^P}, 1\right). \tag{B10}$$

2. V_1^{PIII} Using the substitution $y = C^2$, V_1^{PIII} can be further expressed as follows:

$$\begin{aligned} V_1^{\mathrm{PIII}} = & \frac{1}{2}\int_{C_1^2}^0 \frac{\mathrm{d}y}{(y - C_3^2)\sqrt{[y + (C_1^P)^2][y + (C_2^P)^2]}} \\ & + \frac{1}{2}\int_0^{C_2^2} \frac{\mathrm{d}y}{(y - C_3^2)\sqrt{[y + (C_1^P)^2][y + (C_2^P)^2]}}. \end{aligned} \tag{B11}$$

The evaluation of this integral requires the computation of the integral in the following form:

$$F(x) = \int_{x_0}^{x} \frac{\mathrm{d}t}{(t - I)\sqrt{(t + A)(t + B)}}. \tag{B12}$$

Once $F(x)$ is obtained analytically, the first and second terms of finite integrals in equation (B11) can be evaluated using $\frac{1}{2}(F(0) - F(C_1^2))$ and $\frac{1}{2}(F(C_2^2) - F(0))$, respectively. Note that I in equation (12) is a complex number, in general, and A and B are real. The computation of $G^{\mathrm{P}}(x)$ is discussed in Appendix C.

3. V_1^{PIV} Similar as for V_1^{PII}, use the substitution $C = iC_1^P y$ along Γ_2, and the substitution $C = iC_2^P y$ along Γ_3, we have the following equation:

$$V_1^{\mathrm{PIII}}(a;\Gamma_2) = -\frac{i}{C_2^P C_3}\Pi\left(-\frac{(C_1^P)^2}{(C_3^P)^2}, \frac{C_1^P}{C_2^P}, i\frac{C_1}{C_1^P}\right), \tag{B13}$$

and

$$V_1^{\mathrm{PIII}}(a;\Gamma_3) = -\frac{i}{C_2^P C_3}\Pi\left(-\frac{(C_2^P)^2}{(C_3^P)^2}, \frac{C_2^P}{C_1^P}, 1\right). \tag{B14}$$

APPENDIX C

Computation of $F(x)$

In this section, we consider the indefinite integral:

$$F(x) = \int_{x_0}^{x} \frac{1}{(t-I)\sqrt{(t+A)(t+B)}} \mathrm{d}t. \tag{C1}$$

Because of the symmetry of the integral with respect to A and B, $F(x)$ can be expressed as $F^{(1)}(x)$ or $F^{(2)}(x)$,

$$F^{(1)}(x) = -\frac{2(A+x)}{(I+A)\sqrt{(B+x)(A+x)}} \times \left[{}_2F_1\left(\frac{1}{2}, 1, \frac{3}{2}, \frac{(I+B)(A+x)}{(I+A)(B+x)}\right)\right] + \mathrm{const}_1, \tag{C2}$$

and

$$F^{(2)}(x) = -\frac{2(B+x)}{(I+B)\sqrt{(A+x)(B+x)}} \times \left[{}_2F_1\left(\frac{1}{2}, 1, \frac{3}{2}, \frac{(I+A)(B+x)}{(I+B)(A+x)}\right)\right] + \mathrm{const}_2, \tag{C3}$$

in which ${}_2F_1$ is the Gaussian hypergeometric function, $F^{(1)}(x)$ is not defined at $x = -B$, and $F^{(2)}(x)$ is not defined at $x = -A$. In addition, there are $F^{(1)}(-A) = 0$, $F^{(2)}(-B) = 0$.

Assuming $x_2 > x$ and $A > B$, we then discuss how to use $F^{(1)}$ and $F^{(2)}$ to compute the following definite integral,

$$\int_{x_1}^{x_2} \frac{1}{(t-I)\sqrt{(t+A)(t+B)}} \mathrm{d}t. \tag{C4}$$

- When $x_2 \neq -B$ and $x_1 \neq -A$, either equation (C2) or equation (C3) can be used to compute the definite integral.
- When $x_2 \neq -B$ and $x_1 = -A$, equation (C2) is used to compute the definite integral.
- When $x_2 = -B$ and $x_1 \neq -A$, equation (C3) is used to compute the definite integral.
- When $x_2 = -B$ and $x_1 = -A$, the following equation is used,

$$\int_{-B}^{-A} \frac{1}{(t-I)\sqrt{(t+A)(t+B)}} \mathrm{d}t = -\frac{i}{\sqrt{-(A+I)(B+I)}} \left\{ \ln\left(-i\frac{\sqrt{A-B}\sqrt{B+I}}{\sqrt{-A-I}\sqrt{-A+B}}\right) - \ln\left(-i\frac{\sqrt{A-B}\sqrt{B+I}}{\sqrt{-A-I}\sqrt{-A+B}}\right)\right\}. \tag{C5}$$

In the main text, the case $I = C_3^2$, $A = (C_1^P)^2$, $B = (C_2^P)^2$ is denoted by F^P in equation (18), and the case $I = C_3^2$, $A = (C_1^S)^2$, $B = (C_2^S)^2$ is denoted by F^S in equation (24) and equation (29).

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